Question			Answer	Marks	AOs	Guidance	
8			EITHER Midpoint AB is (3, 1)	**B1**	**3.1a**	Seen or implied	
			Gradient of AB is $\frac{4-(-2)}{7-(-1)}\left(=\frac{6}{8}=\frac{3}{4}\right)$	**M1**	**3.1a**	Must be correct way up – allow one slip	
			Gradient of perpendicular $-\frac{4}{3}$	**M1**	**1.1a**	negative reciprocal FT their gradient of AB	Allow this M1 for showing the perpendicular to the given line is $\frac{1}{2}$.
			Equation of perpendicular bisector $y-1=-\frac{4}{3}(x-3)$	**M1**	**1.1a**	FT their gradient. Condone A or B used instead of M. Do not allow if their gradient of AB used.	
			Solve the simultaneous equations $4x+3y=15$ $2x+y=k$	**M1**	**1.1a**	Attempt to eliminate one variable	
			$y=15-2k$, $x=\frac{1}{2}(3k-15)$	**A1**	**1.1**	cao	
			giving P $\left(\frac{1}{2}(3k-15), 15-2k\right)$	**A1** **[7]**	**1.1**	cao	
			OR P is of the form $(x, k-2x)$	**B1**		Seen or implied	
			and is equidistant from A and B $(x+1)^2+(y+2)^2=(x-7)^2+(y-4)^2$	**M1** **M1**		Finding at least one distance Equating distances	
			$(x+1)^2+(k-2x+2)^2=(x-7)^2+(k-2x-4)^2$ $-8x+1=60-12k$	**M1** **M1**		Substituting for y Attempt to simplify	
			So $x=\frac{1}{2}(3k-15)$ and $y=15-2k$	**A1**		cao	
			P is $\left(\frac{1}{2}(3k-15), 15-2k\right)$	**A1** **[7]**		cao	